Solutions for Discussion Worksheet (Lectures 4 and 5)

September 14, 2025

Note: This section is for instructor use only. Do not distribute to students during the discussion.

1. $P_2(0.1) = 1 + 0.1 + (1/2)(0.1)^2 = 1.105$. $|R_2(0.1)| \le (1/3!) \times 1.11 \times (0.1)^3 \approx 1.85 \times 10^{-4}$.
2. $P_2(t) = 1 - (1/2)t^2$. $\int P_2 = [t - (1/6)t^3]_0^{0.2} \approx 0.2 - 0.001333 \approx 0.198667$. Error bound $(1/3!) \times \max |\sin \xi| \times (0.2)^3 \approx (1/6) \times 0.2 \times 0.008 \approx 2.67 \times 10^{-4}$.
3. Relative error $|3.14159 - 3.1416|/3.14159 \approx 3.18 \times 10^{-6} < 5 \times 10^{-6}$, so 6 significant digits.
4. **Solution 1:** Align: 0.101 + 0.011 = 1.000 (binary, normalized to $1.000 \times 2^{-1} = 0.5$). No tie.

   Solution 2: We are asked to add

   $$1.25 \times 2^{-1} \quad \text{(binary 0.101)}, \qquad 0.75 \times 2^{-1} \quad \text{(binary 0.011)}$$

   in a 6-bit floating point system with 1 sign bit, 3 exponent bits, and 2 fraction bits.

   Step 1. Decimal values.

   $$1.25 \times 2^{-1} = 0.625, \qquad 0.75 \times 2^{-1} = 0.375.$$

   Step 2. Exact addition.

   $$0.625 + 0.375 = 1.000.$$

   Step 3. Normalize.

   $$1.000 = 1.00 \times 2^0.$$

   Step 4. Encode in 6-bit format.

   - Sign bit: 0.
   - Exponent: 0 + bias. With bias = 3, exponent field = 3 = 011.
   - Fraction: 00.

   Thus the bit pattern is

   $$0 \mid 011 \mid 00.$$

   Step 5. Rounding. The result is exactly representable with 2 fraction bits, so no rounding is needed. If a tie had occurred in the third fraction bit, we would apply "round to even," but here this does not arise.

   Final Answer: The sum is 1.0, stored as

   $$0\ 011\ 00$$

   in the 6-bit floating point system.
5. Standard: $\sqrt{40000 - 4} \approx \sqrt{39996} \approx 199.99$. Small root (200 - 199.99)/2 = 0.005. Rationalized: $1/(100 + \sqrt{9999}) \approx 1/(100 + 99.995) \approx 1/199.995 \approx 0.005000$. Relative error standard $\sim$0.02%, rationalized $\sim$0.001%.
6. $P_3(0.1) = 0.1 - (1/6)(0.001) = 0.099833$. $R_3 \le (1/24) \times 1 \times (0.1)^4 = 4.17 \times 10^{-6}$. In 4 digits: 0.1000 - 0.0002 = 0.0998. Total error $\sim 4 \times 10^{-4}$ (round-off dominant).